Graph Theory – Lecture Notes

Sofia Brenner

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Contents

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Layout: Madgdalena Rambau
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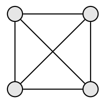
1 Introduction

Definition 1.1. A *(simple) graph* G is a pair (V, E) where V is a non-empty set and $E \subseteq \{\{x, y\} \in V \times V : x \neq y\}$. We call the elements of V *vertices* of G and the elements of E *edges* of G. We also write $V(G)$ for V and $E(G)$ for E, respectively.
'Simple' means no loops and no double edges.

Remark 1.2. V is finite, unless otherwise stated.

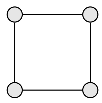
Example 1.3. We define a few common graphs. Let $n, m \in \mathbb{N}$.

- The *complete graph* K_n has n vertices and each pair of vertices is connected.



K_4

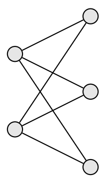
- The *cycle* C_n has n vertices, which are cyclically connected.



C_4

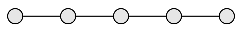
- Let A and B be disjoint sets with $|A| = n$ and $|B| = m$. The *complete bipartite graph* $K_{m,n}$ is defined by

$$V = A \dot{\cup} B \tag{1}$$
$$E = \{\{a, b\} : a \in A, b \in B\} \tag{2}$$



$K_{2,3}$

- The *path* P_n has the vertices $1, \dots, n$ and edges $\{a, a+1\}$ for $a \in \{1, \dots, n-1\}$. (in books sometimes P_{n-1} has n vertices)



P_5

Definition 1.4. Let G be a graph, $u, v \in V(G)$.

- If $\{u, v\} \in E(G)$, then u and v are **adjacent** and the edge $e = \{u, v\}$ is **incident** with both u and v.
- $N_G(v) = \{w \in V(G) : \{v, w\} \in E(G)\}$ is the **neighbourhood** of v in G.
- $\deg_G(v) = |N_G(v)|$ is the **degree** of v.

Lemma 1.5 (Handshake Lemma)**.** *Let G be a graph. Then*

$$\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|. \tag{3}$$

(in particular, the sum of degrees is even).

Proof. $\sum\limits_{v \in V(G)} \deg_G(v) = \{(v, e) : v \in V(G), e \in E(G) \text{ such that } v \text{ incident to } e\} = 2|E(G)|$ □

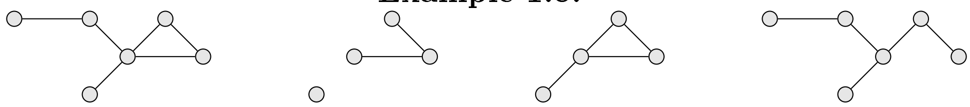
Definition 1.6. Two graphs $G = (V, E)$ and $G' = (V', E')$ are **isomorphic** if there exists an isomorphism ϕ between them, i.e., a bijection $\phi : V \to V'$ such that

$$\forall u, v \in V(G) : \{u, v\} \in E(G) \iff \{\phi(u), \phi(v)\} \in E(G').$$

Definition 1.7. Let $G = (V, E)$ be a graph. A **subgraph** of G is a graph $H = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E \cap \binom{V'}{2}$, where $\binom{S}{K}$ denotes the set of all K-subsets of S.

- H' is an **induced subgraph** of G if it is a subgraph with $E' = E \cap \binom{V'}{2}$.
- H'' is a **spanning subgraph** if it is a subgraph with $V' = V$.
- For $W \subseteq V$, we write $G[W]$ for the induced subgraph on W.

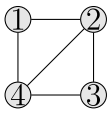
Example 1.8.



G H H' H''

Definition 1.9. Let v_i be vertices of a graph G.

- A **walk** is a sequence of vertices v_i s.t. $\{v_i, v_{i+1}\} \in E(G) \forall i \in \{1, ..., k-1\}$
- The **length** of a walk is k (number of vertices in the walk)
- The walk is **closed** if $v_1 = v_k$
- A **trail** is a walk that does not use an edge twice
- A **path** is a walk for which $v_1, ..., v_k$ are pairwise distinct



- $(1, 2, 3, 4, 2, 3)$ is a walk, but not a trail.
- $(1, 2, 3, 4, 2)$ is a trail, but not a path.
- $(1, 2, 3, 4)$ is a path.

Definition 1.10. Let G be a graph.

- A graph G is **connected** if there is a walk between any pair of its vertices
- A **(connected) component** of G is an inclusion wise maximal connected subgraph

1.1 Hamiltonian Cycles

Definition 1.11. A **Hamiltonian cycle** in a graph G is a spanning subgraph that is a cycle (isomorphic to C_n for some $n \in \mathbb{N}$)
If G has a Hamiltonian cycle, G is called **Hamiltonian**

Theorem 1.12 (Dirac's Theorem)**.** *Let G be a graph of order $n \geq 3$ where the **order** of G is given by $|V(G)|$ and the **size** by $|E(G)|$. If every Vertex of G has degree at least $\frac{n}{2}$ then G is Hamiltonian*

Theorem 1.13 (Ore's theorem)**.** *Let G be a graph of order $n \geq 3$. If for every pair $u, v \in V, u \neq v, u$ and v non-adjacent, we have $\deg(u) + \deg(v) \geq n$ then G is Hamiltonian*

Remark 1.14.
Ore $\implies$ Dirac

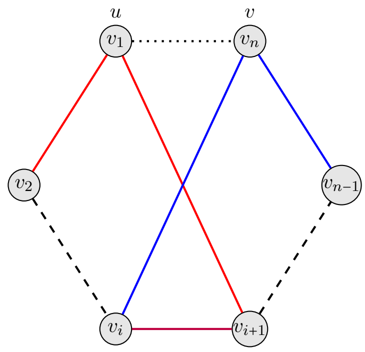
Proof.

The Graph G is connected: suppose G is not connected. Take an arbitrary component of G and call it H. The order of H is $n-k$ if k vertices are not in $V(H)$. Now any $v \in V(H)$ has $\deg(v) \leq n-k-1$ with equality if v is adjacent to every vertex in $V(H)$. Now take $u \notin V(H)$ so $\deg(u) \leq k-1$. It follows that $u, v \in V(G), u \neq v$ and u, v are not adjacent so $\deg(u) + \deg(v) \geq n$ but $\deg(u) + \deg(v) \leq k - 1 + n - k - 1 = n - 2 < n$ ↯

- G has a Hamiltonian cycle: By induction on the number of non-edges
- No non-edge $\implies$ G complete, hence Hamiltonian
- Assume G non complete $u, v \in V, u \neq v$, be non-adjacent. By induction the graph G' obtained from G by adding $e = \{u, v\}$ as an edge is Hamiltonian. Let C be a Hamiltonian cycle of G'. If C does not use $e \implies C$ is Hamiltonian cycle in G. Assume C uses e. Removing e from C yields a path from u to v

  For $i \in \{2, ..., n-1\}$
- If v_i is a neighbour of u, color $\{v_{i-1}, v_i\}$ red
- If v_i is a neighbour of v, color $\{v_i, v_{i+1}\}$ blue

- Since $\deg(u) + \deg(v) \geq n$ and the path $v_1, ..., v_n$ has $n-1$ edges, there exists an edge $\{v_i, v_{i+1}\}$ which is both red and blue. Then $v_1, ...v_i, v_n, v_{n-1}, ..., v_{i+1}$ is a Hamiltonian cycle in G.



□

1.2 Eulerian Trail

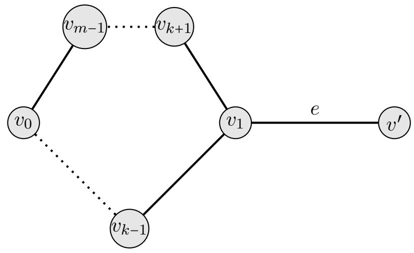
Definition 1.15. An **Eulerian trail** is a trail that passes through each edge of the graph exactly once. A graph with a closed Eulerian trail is called **Eulerian**

Theorem 1.16 (Euler's theorem)**.** *Let G be a connected graph with at least 2 vertices. Then G has a closed Eulerian trail iff every vertex in G has even degree*

Proof.
" $\Longrightarrow$ "
Let T be a closed Eulerian trail and fix an orientation on T. Then for every $v \in V(G)$, the incident edges can be classified as "ingoing" or "outgoing". This defines a bijection between in and outgoing edges of T, so $\deg(v)$ is even.
" $\Longleftarrow$ "
Let $T = (v_0, e_1, v_1, ..., e_m, v_m)$ be a longest trail. We will prove (i) $v_0 = v_m$ and (ii) $\{e_i : i \in \{1, ..., m\}\} = E(G)$. (i)+(ii) $\iff$ T is closed Eulerian trail.

(i) If $v_0 \neq v_m$, then v_0 is incident to an odd number of edges in T. As $\deg(v_0)$ is even, there is $e \in E(G)$ incident to v_0 and not on T. We could thus append e on T ↯ because T is longest trail.

(ii) Claim: $V(T) = V$. Assume otherwise there exists a vertex $v' \notin V(T)$ and $e = \{v_k, v'\} \in E(G)$ for some $k \in \{0, ..., m\}$ (by connectedness). Then $(v', e, v_k, e_{k+1}, v_{k+1}, ..., v_{m-1}, e_m, v_0, e_1, ..., v_{k-1}, e_k)$ is a longer trail hence $V(T) = V$



Claim: $E(T) = E$. Assume otherwise. Consider $e \in E \setminus E(1)$ and write $e = \{v_k, v_l\}$. Then $(v_k, e_{k+1}, v_{k+1}, ..., v_{m-1}, e_m, v_0, e_1, v_1, ..., e_k, v_k, e, v_l)$ is a longer trail ↯

□

Theorem 1.17. *Let G be a connected graph with at least two vertices. Then G has an Eulerian trail iff there are precisely 0 or two vetices of odd degree*

Proof.
By the Euler's Theorem, we know that if there are 0 vertices of odd degree then there exists an Eulerian trail (in fact a closed Eulerian trail), and if there exists a closed Eulerian trail, then all vertices have even degree.

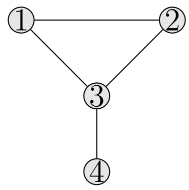
Suppose now that G has exactly two vertices x and y of odd degree. We consider the following graph G': Set $V(G') = V(G) \cup \{z\}$, where $z \notin V(G)$. Furthermore set $E(G') = E(G) \cup \{\{x, z\}, \{y, z\}\}$. Clearly, every vertex has even degree in G', so there exists a closed Eulerian trail. Since z is only adjacent to x and y, these are the neighbours of z on the trail. Deleting the edges $\{x, z\}$ and $\{y, x\}$ from the trail yields an Eulerian trail from x to y in G. Conversely, if there is a non-closed Eulerian trail with endpoints x and y in G, then x and y are non-adjacent. Adding the edge $\{x, y\}$ creates a graph with a closed Eulerian trail, so all vertices have even degree by Euler's theorem. Thus x and y are the only verices in G with odd degree. □

1.3 Datastructures

Assume that $G = (V, E)$ is a graph with $v = \{1, ...n\}$

Definition 1.18. The **adjacency matrix** of G is the matrix $A = (a_{ij})$ with
$$a_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & else \end{cases}$$
The **adjacency list** of G stores a label for each vertex, and a list of (pointers to) its neighbours.

Example 1.19.



$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

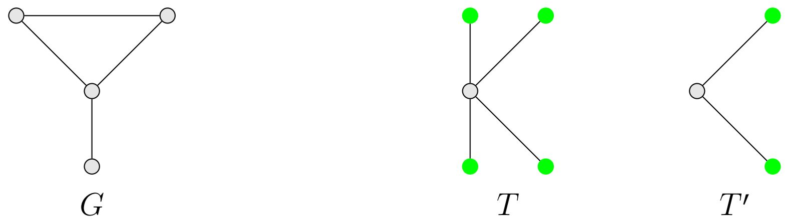
- v_1: v_2, v_4
- v_2: v_1, v_3, v_4
- v_3: v_2
- v_4: v_1, v_2

Remark 1.20.
Memory requirement:
Adjacency matrix: $\theta(n^2)$
Adjacency list: $\theta(n+m)$, where m is the number of edges

2 Trees

Definition 2.1. A **tree** is a graph that is connected and does not contain a cycle (a graph "contains a cycle" if it has a subgraph isomorphic to C_n for some $n \geq 3$).
A **forest** is a graph that does not contain a cycle.
A **leaf** is a vertex of a tree with degree 1.

Example 2.2.



- G is not a tree nor a forest
- T, T' are trees and forests
- $T \cup T'$ is a forest but not a tree
- All green vertices are leafs, all white vertices are not leafs

Lemma 2.3. *Every tree T with $n \geq 2$ vertices has at least one leaf.*

Proof. We iteratively construct a path $v_1, ..., v_n$ as follows:
Start with any edge $\{v_1, v_2\} \in E(T)$. Assume that we have constructed $v_1, ..., v_k$. To construct v_{k+1}, we may assume that v_k is not a leaf in T (or we are done). Otherwise v_k has a neighbour $v_{k+1} \neq v_{k-1}$. $v_{k+1} \neq v_i$ because T is a tree and otherwise we would get a cycle $(v_i, ..., v_k, v_{k+1} = v_k)$. Add v_{k+1} to the path and continue finite many steps (because there are only finitely many vertices). The algorithm just stated ends at a leaf. □

Theorem 2.4. *Let G be a graph with n vertices. The following are equal:*

(i) G is a tree

(ii) G is connected with $n-1$ edges

(iii) G is connected and removing an edge disconnects it

(iv) G has exactly one path between each pair of distinct vertices

(v) G has no cycle, but adding an edge creates a cycle

Proof. "1. $\implies$ 2."
Induction on n:
$n = 1$ is trivial
Let $n \geq 2$. By Lemma 2.3, G has a leaf v. Removing v and the unique edge incident to v yields a tree G'. By induction G' has $n - 2$ edges so G has $n - 1$ edges.

"2. $\implies$ 3."
Suppose there is an edge $e \in E(G)$ and the resulting graph by removing e is still connected. Keep removing as long as the resulting graph is still connected. Let G' be the resulting graph. Observe that G' is a spanning subgraph of G. G' does not contain a cycle as otherwise we could remove one edge from the cycle. By "1. $\implies$ 2.", applied to G', we know that G' has $n-1$ edges $\lightning$ because G had $n-1$ edges and we removed at least one.

"3. $\implies$ 4."
Suppose that there is a pair with $u, v \in V(G)$ s.t. there are two distinct paths u_i, v_i between them. Let i_0 be the maximum index s.t. $u_i = v_i : \forall i \in \{1, ..., i_0\}$. The edge $\{v_{i_0}, v_{i_0+1}\}$ does not lie on the path u_i. The graph obtained by removing $\{v_{i_0}, v_{i_0+1}\}$ is connected as for any walk using this edge we can use u_i to go to u to reach v_{i_0} or v to reach v_{i_0+1}.

"4. $\implies$ 5.
If G had a cycle, then there would exist two paths between each pair of neighbouring vertices on the cycle (clockwise + counter clockwise).
Adding an edge creates a cycle: Let $u, v \in V(G)$ non adjacent. There is one path between the two. Now adding an edge between u, v creates a second path $\lightning$

"5. $\implies$ 1.
We only need to show that G is connected. Suppose u, v are not connected by a walk. Creating an edge between u, v does not result in a cycle because for being isomorphic to C_n one needs at least two walks between each of the vertices in the cycle but by assumption the only walk connecting u, v is the newly added edge $\lightning$ $\square$

Corollary 2.5. *Every connected graph G has a spanning tree.*

Proof.
Consider graphs on n vertices, now do induction on the number of edges m. As G is connected, $m \geq n - 1$. If $m = n - 1 \implies G$ is a tree. If $m > n - 1 \implies G$ is not a

tree. By Theorem 2.4 we can remove edges s.t. the resulting graph is connected. By induction it has a spanning tree, which is then also a spanning tree of G. $\square$

Theorem 2.6 (Matrix-tree theorem). *Let G be a graph with vertices $\{1, ...n\}$, edges $\{e_1, ..., e_m\}$.*

Let $Q = (q_{ij})$ with $q_{ij} = \begin{cases} -1 & \{i,j\} \in E \\ 0 & i \neq j : \{i,j\} \notin E \\ \deg(i) & i = j \end{cases}$ Let Q_{ij} be the matrix obtained from Q by removing the i-th row and the j-th column. Then the number of labelled spanning trees of G is $|\det(Q_{ij})|$, for any choice of i, j.

Corollary 2.7 (Cayley's formula). *The number of labelled spanning trees of K_n is n^{n-2} if $n \geq 2$.*

Proof. Define Q as in Theorem 2.6.

$$Q_{11} = \begin{pmatrix} n-1 & 1 & \cdots & 1 \\ 1 & \ddots & & \vdots \\ \vdots & & \ddots & 1 \\ 1 & \cdots & 1 & n-1 \end{pmatrix} \to \bar{Q}_{11} = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -n & n & 0 & \cdots & 0 \\ -n & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & 0 \\ -n & 0 & \cdots & 0 & n \end{pmatrix} \to \hat{Q}_{11} = \begin{pmatrix} 1 & \star & \cdots & \star \\ 0 & n & & \vdots \\ \vdots & & \ddots & \star \\ 0 & \cdots & 0 & n \end{pmatrix}$$

$$\implies |\det(Q_{11})| = |\det(\hat{Q}_{11})| = n^{n-1-1} = n^{n-2}$$ $\square$

3 Planar Graphs

Definition 3.1. A **planar drawing** of a graph G consists of injective continuous functions

$$\phi_v : V(G) \to \mathbb{R}^2 \qquad \phi_e : [0,1] \to \mathbb{R}^2$$

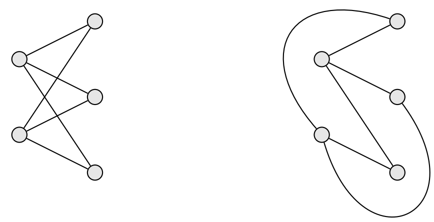
s.t.:

- $\{\phi_{\{u,v\}}(0), \phi_{\{u,v\}}(1)\} = \{\phi_u(u), \phi_v(v)\}$ for all $\{u,v\} \in E(G)$
- The sets $\phi_e((0,1))$ for each $e \in E(G)$ are pairwise disjoint and disjoint from $\mathrm{Im}(\phi_v)$ (i.e. no crossings)

A graph G is called **planar** if it has a planar drawing.
A **plane graph** is a planar graph, together with a fixed planar drawing.

Example 3.2.

- Trees
- K_4
- $K_{2,3}$



Definition 3.3. A **face** of a plane graph G is a maximal connected part of $\mathbb{R}^2 \backslash (\mathrm{Im}(\phi_v(u)) \cup \mathrm{Im}(\phi_e))$. Every plane graph has exactly one unbounded face called the **outer face**. All other faces are called **inner faces**. The **boundary** of a face are the vertices and edges of G incident with it. A plane graph is called **planar triangulation** if every face is bounded by a cycle of length precisely 3 (this includes the outer face).

Theorem 3.4 (Euler's formula)**.** *Let $G = (V, E)$ be a connected planar graph. Then the number of faces $|F|$ of G is $|F| = |E| - |V| + 2$.*

Proof. Induction on $|E|$.
Since G is connected, we have $|E| \geq |V| - 1$ so we start the induction at $|E| = |V| - 1 \implies G$ is a tree. Since a tree does not contain a cycle, we have $|F| = |E| - |V| + 2$. Now let $|E| > |V| - 1$. Then G is not a tree, so G contains a cycle C. Choose an edge e on C and remove it to obtain a plane graph G'. e separates two faces in G, which fuse to a single face in G'. By induction, the number of faces $|F'|$ in G' is $|F| - 1 = |F'| = |E| - 1 - |V| + 2 \implies |F| = |E| - |V| + 2$. □

Lemma 3.5. *For every plane graph $G = (V, E)$ with $|V| \geq 3$, we have $|E| \leq 3|V| - 6$.*

Proof. Wlog G is connected. Consider a planar drawing of G and let F be the set of faces. Every face is bounded by at least 3 edges and each edge bounds at most 2 faces. By Euler's formula 3.4, we obtain $\frac{2}{3}|E| \geq |F| = |E| - |V| + 2 \implies |E| \leq 3|V| - 6$. □

Remark 3.6. This means that the number of edges in a planar graph grows only linear proportional to the number of vertices.

Example 3.7.

- K_5 is not planar (10 edges) due to Theorem 3.4
- $K_{3,3}$ is not planar (without proof)

Definition 3.8. A **subdivision** of a graph G is a graph obtained from G by repeatedly subdividing some of its edges.

Theorem 3.9 (Kuratowski's theorem)**.** *A graph G is planar iff G contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.*

Remark 3.10. A subdivision of a graph is planar iff the graph is planar

Definition 3.11. Let G be a graph. The graph obtained from G by contracting $e = \{u, v\} \in E(G)$ denoted G/e, is the graph obtained from G by deleting u, v, inserting a new vertex w and making w adjacent to all vertices in $N_G(u) \cup N_G(v)$. A graph G' is a **minor** of G if G' can be obtained from G by a sequence of vertex deletions, edge deletions and edge contractions.

Remark 3.12. If G' is a minor of G then G' planar $\Longleftarrow$ G is planar.

Theorem 3.13 (Wagner's theorem)**.** *A graph G is planar iff $K_{3,3}$ and K_5 are not a minor of G.*

4 Flows, Cuts, Connectivity

Definition 4.1. A **directed graph** is a pair (V, E) with $E \subseteq (V \times V) \backslash \{(x, x) : x \in V\}$. Write $\delta_G^+(v)$ for edges starting in v and $\delta_G^-(v)$ for edges ending in v.

Definition 4.2. A **flow network** is a tuple (G, s, t, c) where G is a directed graph, $s, t \in V(G), s \neq t$ and $c : E \to \mathbb{R}_0^+$.

- s is called a **source**
- t is called a **target**
- $\forall e \in E(G) : c(e)$ is the **capacity of** e

A **flow** in a flow network is a function $f : E \to \mathbb{R}_0^+$ s.t.

- $f(e) \leq c(e)$ for all $e \in E(G)$
- $\forall v \notin \{s, t\}, \sum\limits_{e \in \delta^-(v)} f(e) = \sum\limits_{e \in \delta^+(v)} f(e)$ (i.e. what flows in flows out)
- $\forall e \in \delta^-(s) \cup \delta^+(t) : f(e) = 0$ (i.e. nothing flows into a source or out of a target)

The **value** of a flow f is $\mathrm{val}(f) = \sum\limits_{e \in \delta^-(v)} f(e) = \sum\limits_{e \in \delta^+(v)} f(e)$

Remark 4.3. For $v \in V$, write $f(\delta^+(v)) = \sum\limits_{e \in \delta^+(v)} f(e)$; $f(\delta^-(v)) = \sum\limits_{e \in \delta^+(v)} f(e)$

Definition 4.4. A **cut** in a graph/directed graph G is a partition of $V(G)$ into two non-empty parts A and B (i.e. $V(G) = A \dot\cup B$ and $A, B \neq \varnothing$). For $s, t \in V(G), s \neq t \implies (A, B)$ is a (s, t)-cut if $s \in A, t \in B$.
If (S, T) is a (s, t)-cut, write $\delta^+(S) = \{e \in E(G) : e \text{ start in } S \text{ ends in } T\} = \delta^-(T)$ and $\delta^-(S) = \{e \in E(G) : e \text{ starts in } T \text{ and ends in } S\} = \delta^+(T)$.
The **capacity** of the cut is $c(S, T) = \sum\limits_{e \in \delta^+(S)} c(e) = c(\delta^+(S)) = c(\delta^-(T))$.

Theorem 4.5 (max flow-min cut theorem). *Let (G, s, t, c) be a flow network. Then* $\max\{\mathrm{val}(f) : f \textit{ flow}\} = \min\{c(S, T) : (S, T) \textit{ is a } (s, t)\textit{-cut}\}$.

Lemma 4.6. *Let f be a flow, (S,T) be a (s,t)-cut. Then* $\mathrm{val}(f) = f(\delta^+(S)) - f(\delta^-(S))$.

Proof.
Claim: $\forall A \subset V : f(\delta^+(A)) - f(\delta^-(A)) = \sum\limits_{v\in A} f(\delta^+(v)) - f(\delta^-(v)) = (*)$
The sum $f(\delta^+(A)) - f(\delta^-(A))$ does not account for edges $e \in E(G)$ with start and end in A. On the right side, $f(e)$ is counted twice, once with positive and once with negative sign, so the sum vanishes.
Then $\mathrm{val}(f) = f(\delta^-(t)) \overset{(1)}{=} \sum\limits_{v\in T} f(\delta^-(v)) - f(\delta^+(v)) \overset{(*)}{=} f(\delta^-(T)) - f(\delta^+(T)) = f(\delta^+(S)) - f(\delta^-(S))$
(1) holds because $f(\delta^+(t)) = 0$ (t is a sink) and $\forall v \notin \{s,t\} : f(\delta^-(v)) - f(\delta^+(v)) = 0$ (by definition). □

Corollary 4.7. *Let f be a flow, (S,T) an (s,t)-cut.*
$\mathrm{val}(f) \le c(\delta^+(S))$
Thus $\max\{\mathrm{val}(f) : f \textit{ flow}\} \le \min\{c(S,T) : (S,T)\ (s,t)\textit{-cut}\}$.

Proof. Theorem 4.5
max flow $\le$ min cut due to Corollary 4.7
Let f maximum flow (i.e. a flow for which $\mathrm{val}(f)$ is maximal). Let $c' : E(G) \to \mathbb{R}_0^+$ where $c'(e) = c(e) - f(e)$. Let $S \subset V(G)$ be defined as the set of reachable vertices[1] according to the following rules:

- $s \in S$
- $x \in V(G)$ is in S (i.e. reachable) if there is a walk (in the underlying directed graph) from s to x traversing $e_1, ..., e_k$ s.t.
  (i) $c'(e_i) > 0$ ($\iff f(e_i) < c(e_i)$) if e_i is traversed towards x.
  (ii) $c'(e_i) < c(e_i)$ ($\iff f(e_i) > 0$) if e_i is traversed towards s.

If $S = V(G)$, we can improve f as follows:
Let $e_1, \ldots, e_k$ be a walk from s that makes t reachable, and e_i is traversed in orientation $\sigma_i \in \{\pm 1\}$, then $\tilde{f}(e_i) = f(e_i) + \sigma_i\varepsilon$, for $i \in \{1, ..., k\}$ and $\tilde{f}(e_i) = f(e_i)$ otherwise. This defines a flow $\tilde{f}$ with $\mathrm{val}(\tilde{f}) > \mathrm{val}(f)$ for $\varepsilon > 0$ sufficently small $\lightning$
Thus $t \notin S$. Set $T = V \setminus S \neq \varnothing$.

- $\forall e \in \delta^+(S) \quad c'(e) = 0$, i.e. $f(e) = c(e)$
  $e = \{x,y\}$ with $x \in S, y \in T$. If $c'(e) \neq 0 \implies e$ satisfies $(i) \implies y$ reachable $\lightning$

[1] These are the vertices where the flow "is not maximal yet" but has a walk to s.

- $\forall e \in \delta^-(S) \quad c'(e) = c(e) \iff f(e) = 0$

- $e = \{y, x\}$ with $x \in S, y \in T$. If $c'(e) \neq c(e) \implies e$ satisfies $(ii) \implies y$ reachable $\lightning$

- $c(S,T) = \sum_{e \in \delta^+(S)} c(e) = \sum_{e \in \delta^+(S)} f(e) - \sum_{e \in \delta^-(S)} f(e) = f(\delta^+(S)) - f(\delta^-(S)) \overset{(4.6)}{=}$ $\mathrm{val}(f) \implies \mathrm{val}(f) \geq c(S,T) \geq \min(s,t)$-cut □

4.1 Connectivity

Definition 4.8. A graph G is k**-vertex connected** ($k \in \mathbb{N}$) if it has at least $k+1$ vertices and whenever we remove up to $k-1$ vertices, the resulting graph is connected.
A graph G is k**-edge connected** if whenever we remove up to $k-1$ edges, the resulting graph is connected.
The **vertex connectivity (edge connectivity)** is the largest $k \in \mathbb{N}$ s.t. G is k-vertex/edge connected.

Remark 4.9. G graph, k vertex connectivity, λ edge connectivity, δ minimal degree of G, then $k \overset{(i)}{\leq} \lambda \overset{(ii)}{\leq} \delta$

Proof. (i) If we have a set of λ edges whose removal makes the res. graph disconnected, then deleting one endpoint for each edge also disconnects the graph.

(ii) Delete all edges incident to a vertex of minimal degree, this disconnects the graph.

□

Lemma 4.10. *G directed graph, $s,t \in V, s \neq t$. Then the maximal number of edge-disjoint (s,t)-paths (path which do not have any common edges) is equal to the size of a minimal (S,T)-cut .*

Proof. "Max flow-min cut theorem" 4.5 with $\forall e \in E \quad c(e) = 1$. □

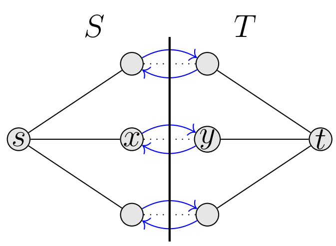
Theorem 4.11 (Menger's theorem for edge-connectivity)**.** *Let G be a graph, $k \in \mathbb{N}$. G is k-edge-connected iff there are k edge-disjoint paths between any pair of vertices in G.*

Proof.
" $\Longleftarrow$ " Removing $k-1$ edges will keep one edge intact, so the resulting graph is connected, hence G is k-edge connected.

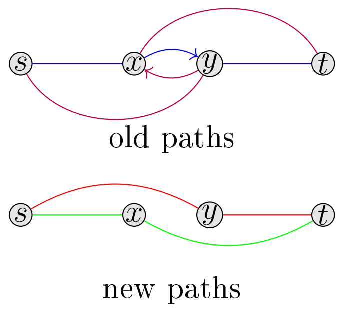
" $\implies$ " Let $s, t \in V, s \neq t$. Let G' be the graph obtained from G by replacing each edge by a double-edge (see graph(1)).

Claim: Any (s,t)-cut in G' has at least size k.
If (S,T) is an (s,t)-cut of size $\leq k-1$, then G has at most $k-1$ edges between S and T. Removing them disconnects the graph $\lightning$ G is k-edge connected. By Lemma 4.10, there are k-edge disjoint paths $p_1, \ldots, p_k$ from s to t in G'.

Suppose $x, y \in V$ s.t. p_i uses $\{x,y\}$ and p_j uses $\{y,x\}$. Replace p_i and p_j by the path following p_i to x and continue via p_j and the path following p_j to y and continue via p_i. Repeat for each such edge. In each step, the sum of the lengths of the paths decreases so the process has to terminate. Then the underlying k-undirected paths in G are edge disjoint (see graph(2)).



graph(1)



graph(2)

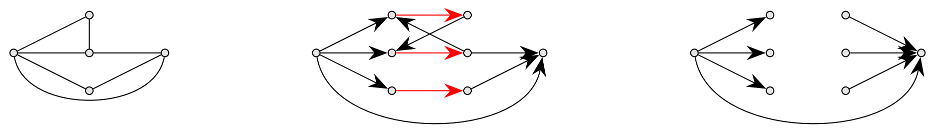
$\square$

Theorem 4.12 (Menger's theorem for vertex-connectivity). *Let G be a graph with at least $k+1$ vertices ($k \in \mathbb{N}$). Then G is k-vertex connected iff there are at least k internally disjoint paths (start and end are the only common vertices) between any pair of distinct vertices of G.*

Proof. " $\impliedby$ " As before (Proof of Theorem 4.11).
" $\implies$ " Let $a, b \in V$ be distinct and $v_1, \ldots, v_n$ be the remaining vertices of G. Consider the following directed Graph G' (see graph(1)):

- $V(G') = \{s, t, v_1', \ldots, v_n', v_1'', \ldots, v_n''\}$
- $E(G') = \{\{v_i', v_i''\} : i \in \{1, \ldots, n\}\} \cup \{\{v_i'', v_j'\} : \{v_i, v_j\} \in E(G)\} \cup \{\{v_j'', v_i'\} : \{v_i, v_j\} \in E(G)\} \cup \{\{s, v_i\} : \{a, v_i\} \in E(G)\} \cup \{\{v_i'', t\} : \{b, v_i\} \in E(G)\} \cup \{\{s,t\} \text{ if } \{a,b\} \in E(G)\}$ where we call elements of $\{\{v_i', v_i''\} : i \in \{1, \ldots, n\}\}$ red edges.

Let k_0 be the minimum size of an (s,t)-cut in G'. If $k_0 \geq k$, G' has k-edge disjoint paths from s to t. Since no red edge is contained in two of those, this corresponds to k internally disjoint paths from a to b in G: each path is of the form $(s, v_1', v_1'', ..., v_m', v_m'', t)$ and corresponds to $(a, v_i, ..., v_m, b)$ in G.
Now assume $k_0 < k$ and consider an (s,t)-cut (S,T) of G' of size k_0.
Claim: We may assume that every edge in $\delta^+(S) \setminus \{(s,t)\}$ is red:
If an edge entering v_i' is in $\delta^+(S)$, move v_i' from T to S without changing the size of the cut. Similarly if an edge leaving v_i'' is in $\delta^+(S)$, move v_i'' to T. If $\{s,t\} \notin E(G)$, then removing the k_0 vertices of G that correspond to red edges in $\delta^+(S)$ disconnects G (no path from s to t in G' after removal $\implies$ none from a to b in G)$\lightning$.
If $\{s,t\} \in E(G)$, proceed as follows:
Let A be the set of vertices in G corresponding to red edges in $\delta^+(S)$ in G' ($|A| = k_0 - 1$). Let $\hat{G}$ be the graph obtained by removing A.
If $\hat{G}$ not connected $\lightning$ G is k-vertex connected.
Thus $\hat{G}$ connected. Since G has $\geq k+1$ vertices either a or b has an additional neighbour.
Wlog a does $\implies$ $\hat{G} - a$ is not connected $\lightning$ G is k-vertex connected (see graph(2)).



G $\qquad$ G' $\qquad$ $\hat{G}$

$\square$

Definition 4.13. A graph is **bipartite** with parts A, B if $V(G) = A \dot{\cup} B$ and every edge in G has one end in A and one in B.

Example 4.14.
- $K_{a,b}$
- $Q_d, d \geq 1$

Definition 4.15. A **matching** in a graph G is a set $M \subset E(G)$ s.t. no two edges in M are incident.
A matching M is **perfect** if every vertex in G is incident to an edge in M.
A maximal matching is a matching where no edge from $E(G)$ can be added to the matching without violating the matching property.

Example 4.16.

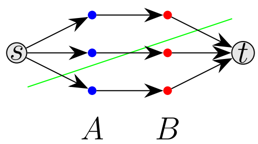
perfect | maximal not perfect | matching not maximal not perfect | not matching

Theorem 4.17 (Hall's marriage theorem)**.** *Let $G = (V, E)$ be a bipartite graph with parts A, B. Then G has a perfect matching iff $\forall P \subseteq A, |N_G(P)| \geq |P|$ and $\forall P \subseteq B, |N_G(P)| \geq |P|$.*

Proof. " $\implies$ " Assume (Wlog) $\exists P \subseteq A$ s.t. $|N_G(P)| < |P|$. Let M be a perfect matching. Then $\forall v \in P$ the other endpoint of the matching edge incident to v is in $N_G(v) \subset N_G(P)$, and these vertices are distinct $\implies |N_G(P)| \geq |P| \lightning$.
" $\impliedby$ " Note that the condition implies $|A| = |B|$, since $|B| \geq |N_G(A)| \geq |A|$ and vice versa. Define an auxiliary graph $G' = (V', E')$ with:

- $V' = V \cup \{s, t\}$
- $E' = \{\{s, a\} : a \in A\} \cup \{\{b, t\} : b \in B\} \cup \{\{a, b\} : a \in A, b \in B\}$
- $\forall e \in E : c(e) = 1$

Every matching M in G defines an integral (s, t)-flow ($f(e) \in \{0, 1\}$) in G with $\operatorname{val}(f) = |M|$ and vice versa.
Let (S, T) be an (s, t)-cut. Let $A_S = A \cap S, A_T = A \cap T, B_S = B \cap S, B_T = B \cap T$.
It follows $|\delta^+(S)| \geq |A_T| + |B_S| + |N_{A_S}(B_T)| \geq |N_{A_T}(B_T)| + |B_S| + |N_{A_S}(B_T)| = |N_A(B_T)| + |B_S| \overset{condition}{\geq} |B_T| + |B_S| = |B| = |A|$.
Thus by maxflow-mincut, $\max\limits_{f \text{ flow}} \operatorname{val}(f) = \max\limits_{f \text{ integral flow}} \operatorname{val}(f) \geq |B|$.
Therefore there exists a matching in G of size $\geq |B|$, i.e. a perfect matching.



∎

5 Coloring

Definition 5.1. Let $k \in \mathbb{N}$. A (proper) k**-vertex-coloring** (k-coloring) of a graph G is a function $c : V(G) \to \{1, 2, \dots, k\}$ such that for every edge $\{u, v\} \in E(G)$, we have $c(u) \neq c(v)$. The **chromatic number** $\chi(G)$ of a graph G is the smallest integer k such that G has a proper k-coloring. We call G k**-colorable** if it has a k-vertex-coloring.
Write $\Delta(G)$ for the maximum degree of a vertex in G.

Lemma 5.2. *Let G be a graph. Then $\chi(G) \leq \Delta(G) + 1$.*

Proof. Order the vertices of G as $v_1, v_2, \dots, v_n$. We will color the vertices in this order. When we come to color v_i, it has at most $\Delta(G)$ neighbors, so there are at most $\Delta(G)$ colors that are forbidden for v_i. Since we have $\Delta(G) + 1$ colors available, there is always at least one color that we can use to color v_i. □

Remark 5.3. The bound $\chi(G) \leq \Delta(G) + 1$ is tight with equality for exactly the complete graphs K_n and the odd cycles C_{2k+1}.

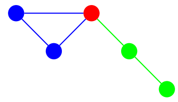
Lemma 5.4. *Let G be a connected graph. If there is $v \in V(G)$ such that* $\deg_G(v) < \Delta(G)$*, then G has a coloring with $\Delta(G)$ colors.*

Proof. Let $w \in V(G)$ such that $\deg_G(w) < \Delta(G)$. Fix a spanning tree T of G rooted at w. Let $v_1, \dots, v_n$ be the vertices of G listed in the way such that each non-root vertex precedes its parent (and the root is the last vertex, i.e., $v_n = w$). Coloar the vertices one by one, starting at v_1. Suppose that $v_1, \dots, v_{i-1}$ are already colored. The vertex v_i is adjacent to at most $\Delta(G) - 1$ vertices among $v_1, \dots, v_{i-1}$: if $i < n$, then the parent of v_i is not among these vertices, and if $i = n$, then $v_i = w$ has degree at most $\Delta(G) - 1$ by assumption. In both cases, there thus exists a color not assigned to any neighbor of v_i, and we can assign it to v_i. This yields vertex coloring of G using at most $\Delta(G)$ colors. □

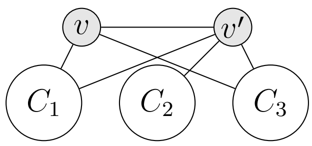
Theorem 5.5 (Brooks' Theorem)**.** *Let G be a connected graph. If G is neither an odd cycle nor a complete graph, then $\chi(G) \leq \Delta(G)$.*

Proof. $\Delta(G) = 0$: Then G is a single vertex and therefore complete.
$\Delta(G) = 1$: Then G is a single edge and therefore complete.
$\Delta(G) = 2$: Then G is a cycle. By assumption, G is an even cycle, so $\chi(G) = 2 = \Delta(G)$.
Let $\Delta(G) \geq 3$
By Lemma 5.4, we may assume that G is $\Delta(G)$-regular ($\forall v \in V(G)$, $\deg_G(v) = \Delta(G)$).

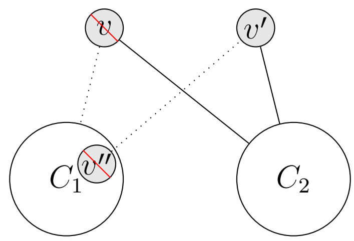
(i) $\exists v \in V(G)$ such that $G-v$ is disconnected. Then $G-v$ has at least two components. Let $C_1, \dots C_k$ be the components of $G-v$, and let $G_i = G[V(C_i) \cup \{v\}]$. G_i is connected, $\Delta(G_i) = \Delta(G)$ and since $\deg_{G_i}(v) < \deg_G(v) = \Delta(G) \implies$ (by Lemma 5.4) we can color G_i with $\Delta(G)$ colors. Fix such a coloring and then permute the colors s.t. v always has the same color. Then this is a coloring of G with $\Delta(G)$ colors.



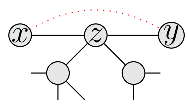
(ii) Wlog G is connected.
Assume not (i) and $\exists v, v' \in V(G)$ s.t. $G - v - v'$ is disconnected.
Let $C'_1, \dots, C'_k$ be the components of $G - v - v'$. Let $G_i = G[V(C'_i) \cup \{v, v'\}]$.
Let G'_i be the graph obtained from G_i by adding $\{v, v'\}$ if it is not already there.
Observe that $\forall i \in \{1, \dots, k\}$, v has a neighbour in C_i (say v'_i).



$$\deg_{G'_i}(v) = \begin{cases} \deg_G(v) - |\bigcup_{j \neq i} N_{C_j}(v)| & \{v, v'\} \in E(G) \\ \deg_G(v) + 1 - |\bigcup_{j \neq i} N_{C_j}(v)| & \{v, v'\} \notin E(G) \end{cases} \leq$$
$$\begin{cases} \Delta(G) - (k-1) & \{v, v'\} \in E(G) \\ \Delta(G) - (k-2) & \{v, v'\} \notin E(G) \end{cases}$$
Same goes for $\deg_{G'_i}(v')$.
If $\deg_{G'_i}(v) < \Delta(G)$ or $\deg_{G'_i}(v') < \Delta(G)$, then we can $\Delta(G)$-color G_i by Lemma 5.4.
If $G'_1, \dots, G'_k$ have a $\Delta(G)$-coloring $\implies$ can combine them to a $\Delta(G)$-coloring of G.
Suppose $\exists i \in \{1, \dots k\}$ s.t. $\deg_{G'_i}(v) = \deg_{G'_i}(v') = \Delta(G)$. Wlog $i = 2$. Then $k = 2, \{v, v'\} \notin E(G)$ ($\deg_{G'_i}(v) < \Delta(G)$ if $k \geq 3$) and $|N_{C_1}(v)| = |N_{C_2}(v')| = 1$.
Let $v'' \in V(C_1)$ be the neighbour of v'
Observation: $\bar{G} = G - v - v''$ is not connected. $\bar{G}$ has components $D_1 := C_1 - v''$ and $D_2 = [V(C_2) \cup \{v'\}]$. v has $\Delta(G) - 1$ neighbours in D_2 and v'' has $\Delta(G) - 1$ neighbours in D_1. Use the above coloring argument on $G - v - v''$.



(iii) Wlog G is connected.
Assume not (i) and not (ii). Then $\forall v, v' \in V(G)$, $G - v - v'$ is connected.
G not complete $\implies \exists x, x' \in V(G), x \neq x'$, non adjacent. Let P be a shortest path from x to x'. Let z be the neighbour of x on P and y be the neighbour of z on P distinct from x. As P is a shortest path, x and y are nonadjacent. Fix a spanning tree of $G - x - y$ rooted at z. Set $v_1 = x, v_2 = y$ and $v_3, ..., v_n$ is a listing of the remaining vertices of G s.t. every vertex in the tree precedes its parents. In particular $v_n = z$. Now we give the same color to v_1 and v_2.
Suppose $v_1, v_2, ..., v_{i-1}$ are colored already.
If $i < n \implies v_i$ is adjacent to at least one vertex in $\{v_{i+1}, ..., v_n\}$ namely its parent. So v_i has at most $\Delta(G) - 1$ neighbours that already have a color, so we have 1 color left.
For $i = n, z$ has two neighbours with same color (x, y), so again there is a color not present among the neighbours of z, which we assign to z.



□

Theorem 5.6 (4-Color Theorem). *Every planar graph is 4-colorable.*

Theorem 5.7 (5-Color Theorem). *Every planar graph is 5-colorable.*

Proof. Induction on $n = |V(G)|$. Fix a plane drawing of G.
$n = 1, 2$: trivial.
Let $n \geq 3$:
Suppose $\exists v \in V(G) \quad \deg_G(v) \leq 4$. Then color $G - v$ by ind. with 5 colors. The neighbours of v use at most 4 colors, so there is a color we can assign to v.
Now assume that G is maximal.
From the proof of Lemma 3.4, it follows that G is a triangulation. By Ex 3.2 $\exists w \in V(G)$ s.t. $\deg_G(w) = 5$. Let $v_1, ..., v_5$ be the neighbours of w, listed in cyclic ordering around w. Then $\{v_1, v_3\}$ or $\{v_2, v_4\}$ is not an edge in $E(G)$, Wlog $\{v_1, v_3\}$.

Let G' be the plane graph obtained by removing w, v_1, v_3 and inserting w' at the position of w. Join every neighbour u of v_1 in G to w' by inserting an edge along $\{u, v_1\}$ and then along $\{v_1, w\}$ (same for the neighbours of v_3 but put only one neighbour of both v_1, v_3). $|V(G')| = n - 2$. G' is planar. By induction, there exists a 5-coloring $c' : V(G') \to \{1, ..., 5\}$ of G'.

Let $c : V(G) \to \{1, ..., 5\}$ with $c(v) = \begin{cases} c'(v), & v \in V(G) \setminus \{w, v_1, v_3\} \\ c(v_1) = c'(w') \\ c(v_3) = c'(w') \\ c(w) \text{ is any color } \notin \{c(v_i) : i = 1, ..., 5\} \end{cases}$

Then c is a 5-coloring of G. $\square$

Definition 5.8. The **clique number** of a graph G, denoted $\omega(G)$, is the order of the largest complete subgraph of G.

Remark 5.9. Clearly, $\omega(G) \le \chi(G) \le \Delta(G) + 1$.

Theorem 5.10. *Let $k \ge 2$. There exists a graph G with $\omega(G) = 2$ and $\chi(G) > k$.*

Definition 5.11. The **girth** of a graph G is the length of the shortest cycle in G.

Theorem 5.12. *For every $k \ge 1$, there exists a graph G with $g(G) > k$ and $\chi(G) > k$.*

Proof. postponed $\square$

Remark 5.13. 5.12 $\implies$ 5.10 as $g(G) > 3 \implies G$ has no triangles $\implies \omega(G) \le 2$.

Conjecture 5.14 (Reed)**.** $\chi(G) \le \lceil \frac{\omega(G)+\Delta(G)+1}{2} \rceil$

5.1 PerfectGraphs

Definition 5.15. A graph G is **perfect** if for every induced subgraph G' of G, we have $\chi(G') = \omega(G')$.

Example 5.16. complete graphs and bipartite graphs are perfect.

Theorem 5.17 (Weak perfect graph theorem)**.** *A graph G is perfect if and only if $\overline{G}$ is perfect.*
($\overline{G}$ is the complement of G which is the graph with the same vertex set as G and two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G)

Theorem 5.18 (Strong perfect graph theorem)**.** *A graph G is perfect if and only if G does not contain an odd cycle of length at least 5 or the complement of an odd cycle of length at least 5 as an induced subgraph.*

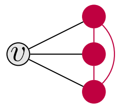
Definition 5.19. A graph is **chordal** if it does not contain an induced cycle of length at least 4.

Remark 5.20. Induced subgraphs of chordal graphs are chordal.

Theorem 5.21. *Every chordal graph is perfect.*

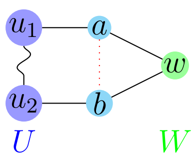
Proof. Use Theorem 5.17. □

Definition 5.22. A vertex v in a graph G is **simplicial** if $N_G(v)$ induces a complete graph.



Lemma 5.23. *Let G be a graph. If G is chordal, then it has a simplicial vertex.*

Proof. Induction on $n := |V(G)|$. Statement is true for $n = 1$ and if G is complete. Now $n \geq 2$ and G not complete.
Choose $U \subseteq V(G)$ of maximal size s.t. $G[U]$(= subgraph induced by the vertices in U) is connected and $U \cup N_G(U) \neq V(G)$ $(N_G(U) = \bigcup_{u \in U} N_G(u) \setminus U)$.
If $u, u' \in V(G)$ are non-adjacent, $u \neq u'$, then $U = \{u\}$ satisfies the above condition.
Let $W = V(G) \setminus (U \cup N_G(U))$, then $W \neq \emptyset$. Every $x \in N_G(U)$ is adjacent to all vertices in W, otherwise we could add x to U $\lightning$ maximality of U. $G[W]$ is chordal and by induction W has a simplicial vertex v.
Claim: v is simplicial in G.
We show that $N_G(U) := N(U)$ is a clique.
Suppose there are $a, b \in N(U)$, a and b non-adjacent, $a \neq b$. Let $w \in W \implies w$ is adjacent to both a and b. Let P be a shortest path from a to b with inner vertices in U (exists since $G[U]$ is connected and both a, b have neihgbours in U).
Then concatenating P with $(b$,$\{b, w\}$,w,$\{a, w\}$,$a)$ yields a cycle in G that does not have a chord, the cycle has length $\geq 4 \lightning G$ chordal.
$\implies N(U)$ is a clique.
$\implies N_G(v) = N(U)$ is a clique $\implies v$ is simplicial in G.



□

Theorem 5.24. *Let G be a graph. Then G is chordal iff G has a perfect elimination ordering, i.e. an ordering $v_1, ..., v_n$ of $V(G)$ s.t. $\forall i \in \{1, \ldots, n\}$, the neighbours of v_i in $\{v_{i+1}, \ldots, v_n\}$ induce a complete graph.*

Proof. " $\Longrightarrow$ " Let v_1 be a simplicial vertex in G (exists by Lemma 5.23). Proceed inductively on $G - v_1$ (which is chordal).
$\Longleftarrow$ Let C be an induced cycle of G of length ≥ 3. Let v_i be a PEO. Let i be minimal s.t. $v_i \in V(C)$. Then the two neighbours of v_i on C are v_{i+j} and v_{i+l} for some $j, l > 0$. Since the neighbours of v_i in $\{v_{i+1}, ..., v_n\}$ induce a complete graph, v_{i+j} and v_{i+l} are adjacent, so $|V(C)| = 3$. Thus G is chordal. □

5.2 EdgeColoring

Definition 5.25. Let $k \in \mathbb{N}$. A (proper) k-**edge-coloring** of a graph G is a function $c : E(G) \to \{1, 2, \ldots, k\}$ such that no two adjacent edges have the same color.
The **chromatic index (edge-chromatic-number)** $\chi'(G)$ of a graph G is the smallest k such that G has a proper k-edge-coloring.

Remark 5.26. $\Delta(G) \leq \chi'(G)$

Theorem 5.27 (Vizing's theorem)**.** *Every graph G has a proper edge coloring with at most $\Delta(G) + 1$ colors. Thus $\chi'(G) \in \{\Delta(G), \Delta(G) + 1\}$.*

Proof. not here. □

6 Ramsey Theory

In this chapter, if we talk about coloring, we always mean a not necessarily proper coloring (think of a child painting the graph randomly).

Example 6.1. In any group of 6 people, there are either 3 people who mutually know or don't know each other. $\iff$ Any Graph with six vertices contains a triangle or an independent set of size 3. $\iff$ Every 2-edge coloring of K_6 contains a monochromatic triangle.

Proof. Let G be a graph with 6 vertices. Pick a vertex v. v has at least 3 neighbours or non-neighbours. Wlog v has 3 neighbours a, b, c. If a, b, c are non-adjacent, we are done. Otherwise, wlog a and b are adjacent, then v, a, b is a triangle. $\square$

Definition 6.2. An **independent set** in a graph G is a set $U \subseteq V(G)$ s.t. $G[U]$ has no edges. Let $\alpha(G)$ denote the **independence number** of G, i.e. the size of the largest independent set in G.

Theorem 6.3 (Pigeonhole Principle)**.** *Let $n_1, ..., n_t \in \mathbb{N}$. Let X be a set with at least $1 + \sum_{i=1}^{t}(n_i - 1)$ elements, and let $X_1, ..., X_t$ be disjoint sets forming a partition of X. Then there exists $i \in \{1, ..., t\}$ s.t. $|X_i| \geq n_i$.*

Proof. Otherwise $|X| = \sum_{i=1}^{t}|X_i| \leq \sum_{i=1}^{t}(n_i - 1) < 1 + \sum_{i=1}^{t}(n_i - 1)$↯ $\square$

Theorem 6.4 (Ramsey's Theorem)**.** *For any $k \geq 2$, $n \in \mathbb{N}$, there exists a $N \in \mathbb{N}$ s.t. every edge coloring of K_N with k colors contains a monochromatic copy of K_n.*

Proof. Set $N = k^{kn}$. Aim: Find $v_1, ..., v_{kn} \in V(K_N)$ s.t. for all edges from v_i to $v_{i+1}, ..., v_{kn}$ all have same color.
We construct W and v_i inductively (v_i are the vertices we want at the end and W is a big enough set of candidates for later iteration steps).

- Set $W_0 := V(K_N)$.
- We want to show that for every $l \in \{1, ..., nk\}$ we can find $\{v_1, ..., v_l\}$ with $\{v_1, ..., v_l\} \cap W_l = \varnothing$ and for every $i = 1, \ldots, l$, the edges from v_i to $W_l \cup \{v_{i+1}, \ldots, v_l\}$ have the same color.

To show that we always find a "next" suitable vertex in W, we look at the following:

- Take $v_{l+1} \in W_l$.

- Of the vertices in $W_l \setminus \{v_{l+1}\}$ at least $\frac{|W_l|-1}{k}$ ($|W_l| - 1$ edges and k colors) are joined to v_{l+1} by edges with the same color (pigeonhole principle 6.3).
- Take this set of vertices as the new W_{l+1}.

Now we show, that we actually have enough vertices in W to do the above construction.
Proof by induction: $|W_j| \geq k^{kn-j} - 1$.
$|W_{l+1}| \geq \frac{|W_l|-1}{k} \geq \frac{k^{kn-l}-1-1}{k} = k^{kn-(l+1)} - 1$.
For $i = 1, \dots, kn$, let c_i be the common color of the edges from v_i to $v_{i+1}, \dots, v_{kn}$. Since there are k colors, there is $I \subseteq \{1, \dots, kn\}$, $I \geq n$ s.t. $c_i = c_j \ \forall i, j \in I$. In particular, the edges in $G[\{v_i : i \in I\}]$ are monochromatic. □

Theorem 6.5. *Let $k, l \in \mathbb{N}$. Let G be a graph of order $\geq \binom{k+l-2}{k-1}$. Then $\omega(G) \geq k$ or $\alpha(G) \geq l$.*

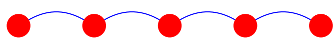
Proof. Like Theorem 6.4. □

Definition 6.6. Let $r(k,l)$ be the minimum N s.t. every graph of order N has $\omega(G) \geq k$ or $\alpha(G) \geq l$. $r(k,l)$ is called the **Ramsey number**. Set $r(n) = r(n,n)$.

Remark 6.7. $r(3,3) = 6$, $r(4,4) = 18$, $43 \leq r(5,5) \leq 48$
$r(4,2) = 4$, $r(4,5) = 25$
The proof of Ramsey's theorem gives $r(n) \leq 4^n$
The best known upper bound is $r(n) \leq 3.9999^n$

Proposition 6.8. $r(n) > (n-1)^2$

Proof. The task is to find a 2-edge coloring of K_N with $N = (n-1)^2$ s.t. there is no monochromatic K_n.
Split $V(K_N)$ into $n-1$ parts of size $n-1$. Color edges between different parts blue and edges within the same part red.
The red graph is a disjoint union of $n-1$ copies of K_{n-1}, so it does not contain a red K_n. Given n vertices, at least two will be in the same part, thus connected by red edge, so there cannot be a blue K_n.



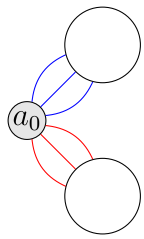
□

Theorem 6.9 (Erdős)**.** $r(n) \geq 2^{\frac{n}{2}}$

Proof. Probabilistic method. Show that a random 2-edge coloring of K_N with $N = 2^{\frac{n}{2}}$ has a positive probability of containing no monochromatic K_n.
Postponed to 7 $\square$

Theorem 6.10. *For every 2-coloring of* $\binom{\mathbb{N}}{2}$*, there exists an infinite set* $X \subseteq \mathbb{N}$ *s.t. all sets in* $\binom{X}{2}$ *have the same color.*

Proof. Find $a_0 \in \mathbb{N}$ and an infinite set $Y_0 \subseteq \mathbb{N} \setminus \{a_0\}$ s.t. all edges $\{a_o, y\}$ with $y \in Y_0$ have the same color. Take $a_1 \in Y_1 \subseteq Y_0$ and proceed analogously $\implies$ find an infinite sequence $a_0, a_1, \ldots$. Because we have only colors red and blue, one of the two needs to repeat infinitely many times in the sequence. $\square$



Remark 6.11. $\mathbb{N}$ can be replaced by any other infinite set.

7 Probabilistic Method

Definition (Rules and key ideas for this chapter)**.** Let $n \in \mathbb{N}$, $V = \{1, ..., n\}$, $\mathcal{G}$ is the set of all graphs with vertex set V. Aim:

- What is the probability that a random graph $G \in \mathcal{G}$ has a certain property $\mathcal{P}$?
- What is the expected value of a certain parameter of a random graph $G \in \mathcal{G}$?

Key Idea: If the probability that a random graph $G \in \mathcal{G}$ has a certain property $\mathcal{P}$ is non zero i.e. positive, then there exists a graph $G \in \mathcal{G}$ with property $\mathcal{P}$.

For every $e \in \binom{V}{2}$, draw the edge with probability $0 \leq p \leq 1$, independently. This will give us a graph G.
If G_0 is a fixed graph, then the probability that $G = G_0$ is $p^m(1-p)^{\binom{n}{2}-m}$ where $m = |E(G_0)|$.

⚠ This does not say anything about isomorphism types.

Definition 7.1. A **(discrete) probability space** is a finite set Ω (usually $\Omega = \mathcal{G}$), together with a probability function $p' : \Omega \to [0,1]$ with the property $\sum\limits_{\omega \in \Omega} p'(\omega) = 1$. An **event** is a subset $A \subseteq \Omega$, the probability of A is $\mathbb{P}(A) = \sum\limits_{\omega \in A} p'(\omega)$.

Here $\Omega = \mathcal{G}$. One needs to show that there exists a probability function $p : \mathcal{G} \to [0,1]$ s.t. each edge occurs independently with probability p.
Let $\mathcal{G}(n,p)$ be the resulting probability space.

Remark. For countable sequences $A_1, A_2, ... \in \Omega$, we have $\mathbb{P}(\bigcup\limits_{i=1}^{\infty} A_i) \leq \sum\limits_{i=1}^{\infty} \mathbb{P}(A_i)$.

Lemma 7.2. *For all $n, k \in \mathbb{N}, n \geq k \geq 2$, the probability that $G \in \mathcal{G}(n,p)$ has an independent set of size $\geq k$ is $\mathbb{P}(\alpha(G) \geq k) \leq \binom{n}{k}(1-p)^{\binom{k}{2}}$.*

Proof. The probability that a fixed set $U \subseteq V$, $|U| = k$, is a independent set in G is $(1-p)^{\binom{k}{2}}$. There are $\binom{n}{k}$ such sets. By Remark 7, we have $\mathbb{P}(\alpha(G) \geq k) \leq \mathbb{P}(\bigcup\limits_{U \subseteq V, |U|=k} \{U \text{ is an independent set}\}) \leq \sum\limits_{U \subseteq V, |U|=k} \mathbb{P}(U \text{ is an independent set}) = \binom{n}{k}(1-p)^{\binom{k}{2}}$.
Analogously, $\mathbb{P}(\omega(G) \geq k) \leq p^{\binom{k}{2}}$. □

Proof to Theorem 6.9. Thm: $r(n) > 2^{\frac{n}{2}}$ for $n \geq 3$.
$n = 3$: $r(3) \geq 3 > 2^{\frac{3}{2}}$.
Let $n \geq 4$: We show that $\forall N \leq 2^{\frac{n}{2}}$ (where N is the order of), the probabilities for $G \in \mathcal{G}(N, \frac{1}{2})$ to have $\omega(G) \geq n$ or $\alpha(G) \geq n$ are both less than $\frac{1}{2}$.
Then $\mathbb{P}(\omega(G) \geq n \text{ or } \alpha(G) \geq n) \leq \mathbb{P}(\omega(G) \geq n) + \mathbb{P}(\alpha(G) \geq n) < \frac{1}{2} + \frac{1}{2} = 1$ so there

exists a graph G on N vertices s.t. $\omega(G) < n$ and $\alpha(G) < n$, thus $r(n) > N$.
$\mathbb{P}(\omega(G) \geq n) \overset{7.2}{\leq} \binom{N}{n}\frac{1}{2}^{\binom{n}{2}} = \frac{N!}{(N-n)!n!}\frac{1}{2}^{\binom{n}{2}} \overset{n!>2}{<} \frac{N!}{(N-n)!2^n}\frac{1}{2}^{\binom{n}{2}} \leq \frac{N^n}{2^n}\frac{1}{2}^{\binom{n}{2}} = \frac{N^n}{2^n}\frac{1}{2}^{\frac{n!}{2(n-2)!}} = \frac{2^{\frac{n}{2}n}}{2^n}2^{\frac{-n(n-1)}{2}} = 2^{\frac{n^2}{2}-n-\frac{n^2}{2}+\frac{n}{2}} = 2^{-\frac{n}{2}} < 1.$ $\square$

Definition. For a probability space (Ω, p) a **random variable** is a function $X : \Omega \to \mathbb{R}$ $[(\Omega, p) = \mathcal{G}(n,p)]$.
Graph invariants like maximal degree, girth, etc. can be interpreted as random variables. The **expection** of X is $\mathbb{E}[X] = \sum_{\omega\in\Omega}\mathbb{P}(\omega)X(\omega)$. Note that $\mathbb{E}[x+y] = \mathbb{E}[x] + \mathbb{E}[y]$ and $\forall\lambda\in\mathbb{R} : \mathbb{E}[\lambda x] = \lambda\mathbb{E}[x]$.

Lemma 7.3. *Let $k \in \mathbb{N}$, $p = p(n)$ a function s.t. $p \geq \frac{6k\ln(n)}{n}$ for large enough n. Let $G \in \mathcal{G}(n, p(n))$. Then $\lim_{n\to\infty}\mathbb{P}(\alpha(G) \geq \frac{1}{2}\frac{n}{k}) = 0$.*

Proof. $\forall n, r \in \mathbb{N}$ with $n \geq r \geq 2$, we have:
$\mathbb{P}[\alpha(G) \geq r] \overset{7.2}{\leq} \binom{n}{r}(1-p)^{\binom{r}{2}} \leq n^r(1-p)^{\binom{r}{2}} = (n(1-p)^{\frac{r-1}{2}})^r \overset{1-p\leq e^{-p}}{\leq} (ne^{-p\frac{r-1}{2}})^r$.
Using $p \geq \frac{6k\ln(n)}{n}$ and $r \geq \frac{1}{2}\frac{n}{k}$, we have

$$ne^{-p\frac{r-1}{2}} = ne^{-\frac{pr}{2}+\frac{p}{2}} \leq ne^{-\frac{3}{2}\ln(n)+\frac{p}{2}} = n^{-1/2}e^{p/2} \leq \sqrt{\frac{e}{n}}.$$

Thus $\lim_{n\to\infty}\mathbb{P}[\alpha(G) \geq \frac{1}{2}\frac{n}{k}] \leq \lim_{n\to\infty}[(ne^{-p\frac{r-1}{2}})^r] \leq \lim_{n\to\infty}[\sqrt{\frac{e}{n}}^{\frac{n}{2k}}] = 0$. $\square$

Lemma 7.4. *Fix n, p and $k \geq 3$. Let $X : \mathcal{G}(n,p) \to \mathbb{N}$ be the number of k-cycles. The expected number of k-cycles is $\mathbb{E}[X] = \frac{(n)_k}{2k}p^k$, where $(n)_k = n(n-1)...(n-k+1)$.*

Proof. For every possible k-cycle C with vertices in $\{1, ..., n\}$, let $X_C : \mathcal{G}(n,p) \to \{0,1\}, G \to \begin{cases} 1, & C \subseteq G \\ 0, & \text{else} \end{cases}$.
$\mathbb{E}[X_C] = \mathbb{P}(X_C = 1) = \mathbb{P}(C \subseteq G) = p^k$. There are $(n)_k$ sequences $v_1, ..., v_k$ of distinct vertices in $\{1, ..., n\}$, but each cycle is labeled by $2k$ such sequences so there are $\frac{(n)_k}{2k}$ possible cycles.
$X = \sum_{C\ k\text{-cycle}} X_C$ so $\mathbb{E}[X] = \sum_{C\ k\text{-cycle}} \mathbb{E}[X_C] = \frac{(n)_k}{2k}p^k$. $\square$

Lemma 7.5 (Markov's inequality)**.** *Let $X : \Omega \to \mathbb{R}$. Then $\forall a > 0 : \mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$.*

Theorem 5.12. Thm: For every $k \in \mathbb{N}$, there exists a graph with girth $g(G) > k$ and chromatic number $\chi(G) > k$.

We may assume $k \geq 3$.
Fix ε with $0 < \varepsilon < \frac{1}{k}$, $p = p(n) = n^{\varepsilon-1}$.

Let $X : \mathcal{G}(n,p) \to \mathbb{R}$ be a random variable assigning to every graph $G \in \mathcal{G}(n,p)$ the number of cycles of length $\leq k$.
Want to enforce $\mathbb{P}[X \geq \frac{n}{2}] < \frac{1}{2}$ (few small cycles), $\mathbb{P}[\alpha(G) \geq \frac{1}{2}\frac{n}{k}] < \frac{1}{2}$ (few independent sets).
By Lemma 7.4, $\mathbb{E}[X] = \sum\limits_{i=3}^{k} \frac{(n)_i}{2i} p^i \leq \frac{1}{2} \sum\limits_{i=3}^{k} n^i p^i \leq \frac{1}{2}(k-2) n^k p^k$.
By Lemma 7.5, $\mathbb{P}[X \geq \frac{n}{2}] \leq \frac{\mathbb{E}[X]}{\frac{n}{2}} \leq (k-2) n^{k-1} p^k \overset{np=n^\varepsilon}{=} (k-2) n^{k-1} n^{k(\varepsilon-1)} = (k-2) n^{k-1+k\varepsilon-k} = (k-2) n^{k\varepsilon-1}$.
As $k\varepsilon - 1 < 0$, this yields $\lim\limits_{n\to\infty} \mathbb{P}[X \geq \frac{n}{2}] = 0$. By choosing n sufficiently large, we can enforce $\mathbb{P}[X \geq \frac{n}{2}] < \frac{1}{2}$ and $\mathbb{P}[\alpha(G) \geq \frac{1}{2}\frac{n}{k}] < \frac{1}{2}$ by Lemma 7.3.
Then $\mathbb{P}[X \geq \frac{n}{2}$ or $\alpha(G) \geq \frac{1}{2}\frac{n}{k}] \leq \mathbb{P}[X \geq \frac{n}{2}] + \mathbb{P}[\alpha(G) \geq \frac{1}{2}\frac{n}{k}] < \frac{1}{2} + \frac{1}{2} = 1 \implies$ There exists a graph G with $< \frac{n}{2}$ cycles of length $\leq k$ and $\alpha(G) < \frac{1}{2}\frac{n}{k}$.
Now delete a vertex from each cycle of length $\leq k$, call the resulting graph H. Then $|V(H)| \geq \frac{n}{2}$, $g(H) > k$ and thus $\chi(H) \geq \frac{|V(H)|}{\alpha(H)} > \frac{\frac{n}{2}}{\frac{1}{2}\frac{n}{k}} = k$. □

Definition 7.6. Fix a probability space. Events A, B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, which is equivalent to $\mathbb{P}(A|B) = \mathbb{P}(A)$ provided that $\mathbb{P}(B) \neq 0$ (where $\mathbb{P}(A|B) = \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}$).
An event is **mutually independent** from events $B_1, ..., B_n$ if for every $I \subseteq \{1, ..., n\}$: $\mathbb{P}(A | \bigcap_{i\in I} B_i) = \mathbb{P}(A)$.
A **dependency graph** for events $B_1, ..., B_n$ is a graph on a vertex set $\{B_1, ..., B_n\}$ and a set of edges s.t. $\forall i \in \{1, ..., n\}$, the event B_i is mutually independent from $V \setminus (\{B_i\} \cup N(B_i))$.

Theorem 7.7 (Lovász Local Lemma)**.** *Let $B_1, ..., B_n$ be events in a (finite) probability space. If there exists $p \in \mathbb{R}$ with $0 \leq p < 1$ s.t.*

- $\mathbb{P}(B_i) < p$ *for all* $i \in \{1, ..., n\}$
- *the max. degree in the dependency graph for $B_1, ..., B_n$ is at most d*
- $p \leq \frac{1}{4d}$ *(d is at least 1 (graph connected) so $p \leq \frac{1}{4}$)*

Then $\mathbb{P}(\bigcap\limits_{i=1}^{n} \overline{B_i}) > 0$ ($\overline{B}$ is the complement of B).

Proof. For $S \subseteq \{1, ..., n\}$, we show inductively that:

(i) $\mathbb{P}(\bigcap\limits_{i\in S} \overline{B_i}) > 0$

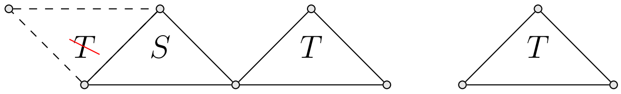
(ii) $\forall k \in \{1, ..., n\} : \mathbb{P}(B_k | \bigcap\limits_{i\in S} \overline{B_i}) \leq 2p$

Wlog assume that the dependency graph is connected.
Proof by induction on $|S|$.
$|S| = 1$, i.e. $S = \{k\}$: (i):$\mathbb{P}(\overline{B_k}) = 1 - \mathbb{P}(B_k) > 1 - p > 0$.
(ii): If B_i and B_k are mutually independent, then $\mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) = \mathbb{P}(B_k) \leq p \leq 2p$.
If B_i and B_k are not mutually independent, then $\mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) = \frac{\mathbb{P}(B_k \cap \overline{B_i})}{\mathbb{P}(\overline{B_i})} \leq \frac{\mathbb{P}(B_k)}{\mathbb{P}(\bigcap_{i \in S} \overline{B_i})} \overset{(i)}{\leq}$
$\frac{p}{1-p} \leq 2p$.
Wlog $S = \{1, ..., s\}$
(i): $\mathbb{P}(\bigcap_{i=1}^{s} \overline{B_i}) = \mathbb{P}(\overline{B}_1)\mathbb{P}(\overline{B_2}|\overline{B_1})...\mathbb{P}(\overline{B_s}| \bigcap_{i=1}^{s-1} \overline{B_i}) = \prod_{i=1}^{s} \mathbb{P}(\overline{B_i}| \bigcap_{j=1}^{i-1} \overline{B_j}) = \prod_{i=1}^{s}(1 - \mathbb{P}(B_i | \bigcap_{j=1}^{i-1} \overline{B_j})) \overset{HI}{\geq}$
$s(1 - 2p) > 0$.
(ii): Need to show that $\mathbb{P}(B_k | \bigcap_{i=1}^{s} \overline{B_i}) \leq 2p$ ($k \in \{1, ..., n\}$).
We split $S = S_1 \cup S_2$ s.t. B_k is mutually independent from S_2. If $S_2 = S$, then $\mathbb{P}(B_k | \bigcap_{i \in S} \overline{B_i}) = \mathbb{P}(B_k) < p$. So assume $S_2 \subsetneq S$.
Set $F_S = \bigcap_{i \in S} \overline{B_i} \overset{(1)}{\neq} \emptyset$, $F_{S_1} = \bigcap_{i \in S_1} \overline{B_i}$, $F_{S_2} = \bigcap_{i \in S_2} \overline{B_i}$.
We want to show that $\mathbb{P}(B_k|F_S) \leq 2p$.
We have: $\mathbb{P}(B_k|F_S) = \frac{\mathbb{P}(B_k \cap F_S)}{\mathbb{P}(F_S)} = \frac{\mathbb{P}(B_k \cap F_{S_1}|F_{S_2})\mathbb{P}(F_{S_2})}{\mathbb{P}(F_{S_1}|F_{S_2})\mathbb{P}(F_{S_2})} = \frac{\mathbb{P}(B_k \cap F_{S_1}|F_{S_2})}{\mathbb{P}(F_{S_1}|F_{S_2})} = (*)$
We have: $\mathbb{P}(B_k \cap F_{S_1}|F_{S_2}) \leq \mathbb{P}(B_k|F_{S_2}) \leq \mathbb{P}(B_k) \leq p$.
For the denominator, we calculate: $\mathbb{P}(F_{S_1}|F_{S_2}) = 1 - \mathbb{P}(\bigcup_{i \in S_1} B_i|F_{S_2}) \geq 1 - \sum_{i \in S_1} \mathbb{P}(B_i|F_{S_2}) \overset{IH}{\geq}$
$1 - \sum_{i \in S_1} 2p \overset{S_2 \subsetneq S}{\geq} 1 - 2pd \overset{p \leq \frac{1}{4d}}{\geq} \frac{1}{2}$.
$\implies (*) \leq \frac{p}{\frac{1}{2}} = 2p$. □

Remark 7.8.
- The bounds can be slightly improved
- Asymmetric version exists
- There is an algorithmic version, the Moser-Tardos-Algorithm

Lemma 7.9. *Let $k, n \in \mathbb{N}_{\geq 2}$ s.t. $4\binom{k}{2}\binom{n-k}{k-2}2^{1-\binom{k}{2}} < 1$.*
Then there exists a 2-edge coloring of K_n that does not contain a monochromatic K_k. Thus $r(k) > n$.

Proof. Consider a random 2-edge coloring of K_n. For $S \subseteq \{1, ..., n\}$ of size k, consider the event A_S that the graph induced by S is monochromatic. Thus $\mathbb{P}(A_S) = \frac{2}{2^{\binom{k}{2}}} = 2^{1-\binom{k}{2}}$. Set $p = 2^{1-\binom{k}{2}} + \varepsilon$ with $\varepsilon > 0$.
A_S is mutually independent from the events A_T with $|S \cap T| \leq 1$.



Thus the maximal degree of the correspondingdependency graph satisfies $d \leq \binom{k}{2}\binom{n-2}{k-2}$.
Then $p \leq \frac{1}{4d}$. Thus by Theorem 7.7 we have $\mathbb{P}(\bigcap_{S\subseteq\{1,\ldots,n\},|S|=k} \overline{A_S}) > 0$ ($\mathbb{P}$(no monochromaticK_k) > 0).
Thus there exists a 2-edge coloring of K_n that does not contain a monochromatic K_k. $\square$

Remark 7.10. This yields $r(k) > \frac{\sqrt{2}}{4}(1 + o(1))k2^{\frac{k}{2}}$.

8 Tree Decompositions And Treewidth

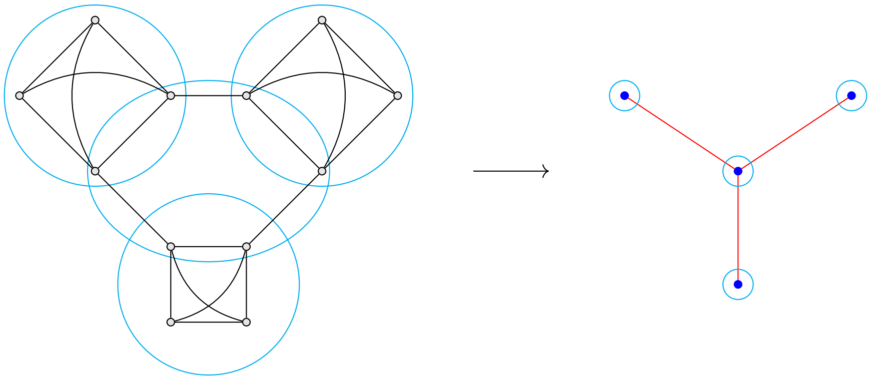
Definition 8.1. A **tree decomposition** of a graph $G = (V, E)$ is a pair $D = (S, T)$, where $S = \{X_i : i \in I\}$ is a set of subsets of V and $T = (I, F)$ is a tree such that the following hold:

(i) $\bigcup_{i \in I} X_i = V$;

(ii) For every edge $\{u, v\} \in E$, there is an $i \in I$ such that $u, v \in X_i$;

(iii) For all $v \in V$, the subtree of T induced by $\{i \in I : v \in X_i\}$ is connected;

The sets X_i are called **bags**.
The **width** of D is $wd(D) = \max_{i \in I} |X_i| - 1$.
The **treewidth** of G is $tw(G) = \min\{wd(D) : D \text{ is a tree decomposition of } G\}$.



Remark 8.2. Every graph has a trivial tree decomposition of width $|V(G)| - 1$, which has a single bag containing all vertices of G.

Remark 8.3. Equivalent formulations for (iii):

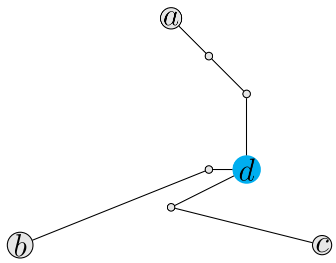
(iii') If, for $v \in V$, we have $v \in X_i$ and $v \in X_j$, then $v \in X_k$ for all $k \in I$ that lie on the unique path from i to j in T.

(iii") For $i, j \in I$ and $k \in I$ on the unique path from i to j, $X_i \cap X_j \subseteq X_k$.

Since the treewidth of a graph G is the maximum of the treewidths of its connected components, we will focus on connected graphs. We have $G = K_1 \iff tw(G) = 0$, we assume $|V(G)| \geq 2$ from now on.

Lemma 8.4. *Let G be a connected graph with tree decomposition $D = (S, T)$. For every clique $C \subseteq V(G)$, there exists a bag X_i, with $C \subseteq X_i$.*

Proof. Induction on $k = |C|$. For $k = 1$, this is (i), for $K = 2$ it's (ii) (Definition 8.1). Now let $k \geq 3$ and let u, v, w be distinct vertices in C. By IH, there exist bags X_a, X_b, X_c s.t. $C \setminus \{u\} \subseteq X_a$, $\quad C \setminus \{v\} \subseteq X_b$, $\quad C \setminus \{w\} \subseteq X_c$. If any two of a, b, c are equal, the corresponding bag contains C. Thus let a, b, c be pairwise distinct. Let P_{ab}, P_{ac}, P_{bc} be the unique paths between ab, ac, bc in T. They must share a vertex $d \notin \{a, b, c\}$ since concatenating P_{ab}, P_{ac}, P_{bc} would be a cycle otherwise, but T is a tree.



By (iii'') from 8.3, we have $C \setminus \{u, v\} \subseteq X_a \cap X_b \subseteq X_d$.
Similarly, $C \setminus \{u, w\} \subseteq X_d$ and $C \setminus \{v, w\} \subseteq X_d \implies C \subseteq X_d$. □

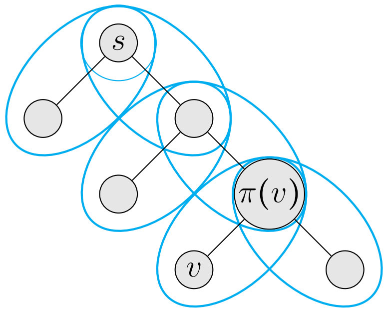
Corollary 8.5. *For every graph G, we have $\omega(G) - 1 \leq tw(G)$.*

Lemma 8.6. *If H is a minor of G, then $tw(H) \leq tw(G)$*

Proof. If H is obtained from G by deleting a vertex or an edge, then $tw(H) \leq tw(G)$ as every tree decomposition of G remains one for H (after deleting the vertex/vertices from all bags containing them).
Suppose H is obtained from G by contracting an edge $\{u, v\}$.
Let $D = (S, T)$ be a tree decomposition of G. Replace all occurences of u and v in the bags by a new vertex w. Since bags containing u induce a subtree of T, similarly for v, and there is a bag containing u and v, the bag containing w induces a subtree in T as well. □

Lemma 8.7. *Let $G = (V, E)$ connected with $|V| \geq 2$. Then $tw(G) = 1 \iff G$ is a tree.*

Proof. " $\Longleftarrow$ "
Let G be a tree. Choose $s \in V$ and run DFS (Depth-First Search) on G. This yields for every $v \in V \setminus \{s\}$ a unique predecessor $\pi(v)$.
Set $X_s = \{s\}$, $X_v = \{v, \pi(v)\}$ $\forall v \in V \setminus \{s\}$, $T = G$, $S = \{X_v : v \in V\}$.
We claim that $D = (T, S)$ is a tree decomposition of G.
Properties (i) and (ii) directly follow from the constructions. For $v \in V$, the set of bags containing v is X_v together with all X_w where w is a child of v. Thus the subtree induced by the corresponding vertices in T is connected. $\implies tw(G) = 1$ (bags only have a max. of 2 vertices in them).



" $\implies$ "
If G is not a tree $\implies$ G contains a cycle and hence a K_3 minor. By Corollary 8.5, we have $tw(K_3) = 2$ and by Lemma 8.6, $2 = tw(K_3) \leq tw(G)$ $\lightning$ $\implies$ G is a tree. $\square$

Remark. Computing treewidth is NP-hard

Theorem 8.8 (Bodlaender). *There exists an algorithm that, given a graph G with $tw(G) = k$, computes a tree decomposition of G of width k in time:* $2^{\mathcal{O}(k^3)}|V(G)|$

Remark.
- Bodlaender et al. (2016): An algorithm that constructs a treedecomposition of width $\leq 5k+4$ or decides $tw(G) \geq k$ in $\mathcal{O}(2^{\mathcal{O}(k)}n)$
- Korhonen (2021): width $\leq 2k+1$ with same runtime
- Korhonen, Lokshtanov (2023): width k, runtime $2^{\mathcal{O}(k^2)}\mathcal{O}(n^4)$
- open question: What is the complexity of computing treewidth on planar graphs. Is it in P?

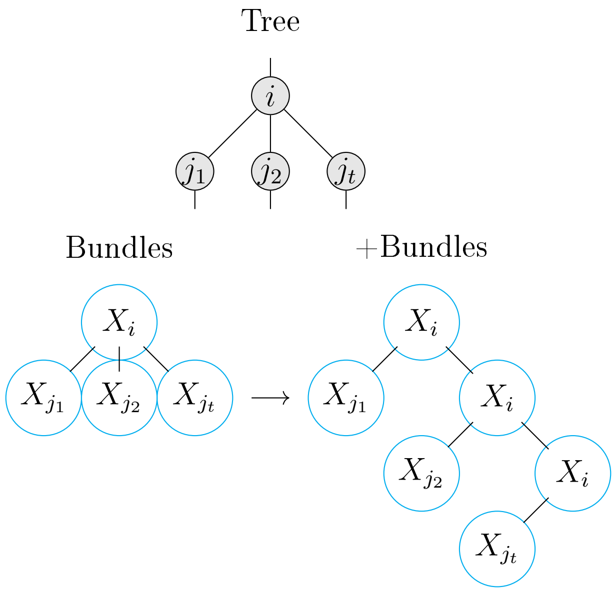
Definition 8.9. A **nice treedecomposition** $D = (S,T)$ of a graph $G = (V,E)$ is a tree decomposition in which a root vertex can be choosen s.t. T is a binary tree (i.e. every vertex has at most 2 children) and such that every vertex of T is of one of the following types:

- **leaf**: if $i \in V(T)$ is a leaf of T, then $|X_i| = 1$
- **introduce node**: if $i \in V(T)$ has exactly one child j and $X_i = X_j \cap \{v\}$ for some $v \in V(G)$
- **forget node**: if $i \in V(T)$ has exactly one child j and $X_i = X_j \setminus \{v\}$ for some $v \in X_j$
- **join node**: if $i \in V(T)$ has precisely two children j_1, j_2 and $X_i = X_{j_1} = X_{j_2}$

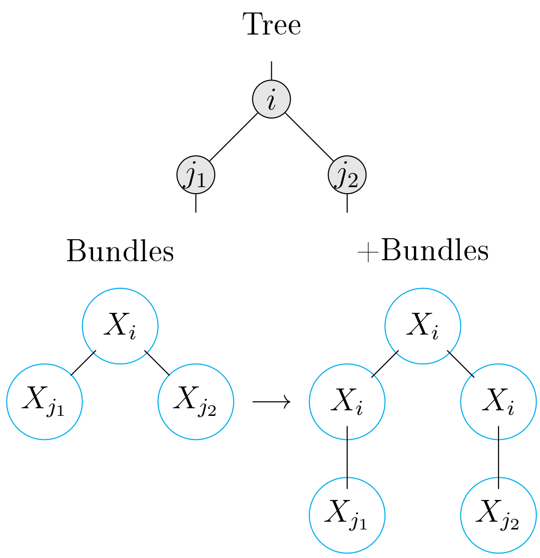
Theorem 8.10. *Let $D = (S,T)$ a tree decomposition of $G = (V,E)$. Then we can construct a nice treedecomposition $D' = (S',T')$ of G with $tw(D') \leq tw(D)$, $|V(T')| \leq ctw(D)|V(T)|$ in polynomial time.*

Proof. Root T at an arbitrary vertex $s \in V(T)$.

(i) Make T binary:
For every $i \in V(T)$ that has ≥ 3 children $j_1, \ldots, j_t$, replace i and its descendants as follows:

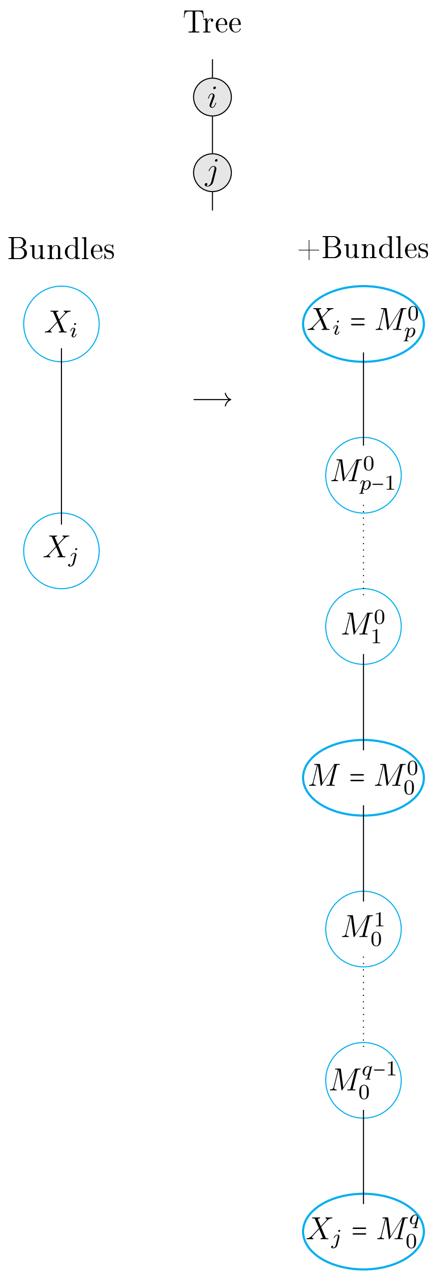


(ii) Make all vertices with precisely two children join nodes:

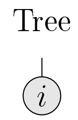


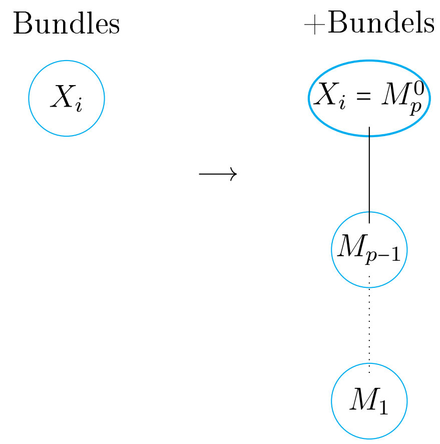
(iii) Make vertices with precisely one child introduce/forget nodes:
Let $i \in V(T)$ with a single child j. Set $M = X_i \cap X_j$ and write $X_i =$

$M\dot\cup\{v_1,...,v_p\}$, $X_j = M\dot\cup\{u_1,...,u_q\}$ with $\{v_1,...,v_p\}\cap\{u_1,...,u_q\} = \varnothing$. Now add bags $M_a^b = M\dot\cup\{v_1,...,v_a\}\cup\{u_1,...,u_b\}$.
Replace the edge $\{i,j\}$ by M_a^b for $a\in\{0,...,p\}, b = 0$ and $b\in\{1,...,q\}, a = 0$.
Connect M_a^b with M_{a-1}^b if $a\neq 0$ and to M_a^{b+1} if $a = 0$.



(iv) Leafs: Let $i\in V(T)$ be a leaf. Write $X_i = \{v_1,...,v_p\} = M_p$

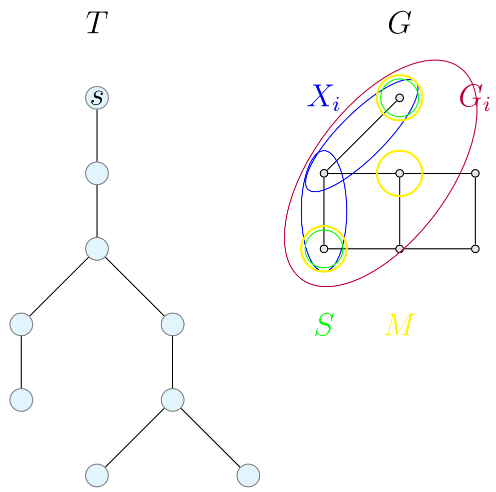




This yields a nice tree decomposition D' with $tw(D)|V(T)|c$ vertices and $tw(D') \leq tw(D)$. $\square$

Lemma 8.11. *Let $G = (V, E)$ be a graph and let $D = (S, T)$ be a nice treedecomposition of $tw(D) = k$. An independent set in G of size $\alpha(G)$ can be computed in $\mathcal{O}(k2^{k+1}n)$, where $n = |V(G)|$.*

Proof. For $i \in V(T)$, let $V_i = X_i \cup \bigcup_{j \text{ descendent of } i \text{ in } T} X_j$. Set $G_i = G[V_i]$.



For all $i \in V(T)$ and $S \subseteq X_i$, let $C_i(S) = \max\{|M| : M$ independent set in $G_i, M \cap X_i = S\}$. Set $C_i(S) = -\infty$ if the above set is empty.
For every $i \in V(T)$ make a table C_i containing $C_i(S) \quad \forall S \subseteq X_i$. Then C_i has $2^{|X_i|} \leq 2^{k+1}$ entries.
If we manage to compute $C_i(S)$ for all i and S then $\alpha(G) = \max\{|M| : \text{independent set in } G\} = \max\{C_s(S) : S \subseteq X_s\}$ with s root of T.
Note $G_s = G$. This can be computed in $\mathcal{O}(2^{k+1})$.
Now compute $C_i(S)$ starting from leafs of T.

(i) $i \in V(T)$ leaf: $X_i = \{v\}$, $G_i = G[V_i] = G[\{v\}] = \{v\}$. $S = \varnothing \implies C_i(\varnothing) = 0$, $S = \{v\} \implies C_i(S) = C_i(\{v\}) = 1$.

(ii) $i \in V(T)$ introduce node: Let j be the unique child of i in T, $X_i = X_j \cup \{v\}$.
Let $S \subseteq X_i = X_j \cup \{v\}$. If $v \notin S$, $C_i(S) = C_j(S)$.
If $v \in S$, then $C_i(S) = \begin{cases} -\infty & \text{if } \{v,w\} \in E \text{ for some } w \in S \\ C_j(S) + 1 & if \{v,w\} \notin E \text{ for all } w \in S \end{cases}$ since v is not adjacent to any vertex of $V_j \setminus X_j$ by Property (iii) 8.1.
If we know C_j, we can compute C_i in $\mathcal{O}(k2^{k+1})$.

□

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